

```

#include <Servo.h>
#define CUSTOM_SETTINGS
#define INCLUDE_GAMEPAD_MODULE
#include <Dabble.h>
#include <SoftwareSerial.h>

SoftwareSerial BT(9,10);

// ===== Servos =====
Servo L1C,L1F,L1T;
Servo L2C,L2F,L2T;
Servo L3C,L3F,L3T;
Servo L4C,L4F,L4T;

// Ultrasonic
const int trigPin = A5;
const int echoPin = A4;

bool obstacleMode = false;

// =====
// DISTANCE FUNCTION
// =====
float getDistance() {
    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);

    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);
    digitalWrite(trigPin, LOW);

    long duration = pulseIn(echoPin, HIGH);

    float distance = duration * 0.0343 / 2.0;
    Serial.println(distance);
    return distance;
}

// =====

```

```
// YOUR FUNCTIONS HERE
// =====
```

```
void stand() {
    // Coxa
    L1C.write(70);
    L2C.write(110);
    L3C.write(110);
    L4C.write(70);

    // Femur
    L1F.write(70);
    L2F.write(120);
    L3F.write(120);
    L4F.write(70);

    // Tibia
    L1T.write(60);
    L2T.write(120);
    L3T.write(120);
    L4T.write(60);

    delay(500);
}
```

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// =====
// FORWARD
// =====
```

```
void forwardStep()
{
    // Lift L1 + L4
    L1F.write(50);
    L4F.write(50);
    delay(50);

    // Move forward
    L1C.write(55);
    L4C.write(85);
}
```

```

    delay(50);

    // Put down
    L1F.write(70);
    L4F.write(70);
    delay(50);

    // Lift L2 + L3
    L2F.write(140);
    L3F.write(140);
    delay(50);

    // Push body forward
    L1C.write(70);
    L4C.write(70);
    L2C.write(125);
    L3C.write(95);
    delay(50);

    // Put down
    L2F.write(120);
    L3F.write(120);
    delay(50);

    // Return
    L2C.write(110);
    L3C.write(110);
    delay(50);
}
// =====
// BACKWARD
// =====
void backwardStep()
{
    L1F.write(50);
    L4F.write(50);
    delay(50);

    L1C.write(85);

```

```

    L4C.write(55);
    delay(50);

    L1F.write(70);
    L4F.write(70);
    delay(50);

    L2F.write(140);
    L3F.write(140);
    delay(50);

    L1C.write(70);
    L4C.write(70);
    L2C.write(95);
    L3C.write(125);
    delay(50);

    L2F.write(120);
    L3F.write(120);
    delay(50);

    L2C.write(110);
    L3C.write(110);
    delay(50);
}
// =====
// TURN LEFT
// =====
void turnLeft() {
    // Lift diagonal pair
    L1F.write(50);
    L4F.write(50);
    delay(50);

    // Left side backward, right side forward
    L1C.write(85);
    L4C.write(85);

    delay(80);

```

```

    L1F.write(70);
    L4F.write(70);
    delay(50);

    // Other diagonal pair
    L2F.write(140);
    L3F.write(140);
    delay(50);

    L2C.write(125);
    L3C.write(125);

    delay(80);

    L2F.write(120);
    L3F.write(120);
    delay(50);

    stand();
}

// =====
// TURN RIGHT
// =====
void turnRight() {
    // Lift diagonal pair
    L1F.write(50);
    L4F.write(50);
    delay(50);

    // Left side forward, right side backward
    L1C.write(55);
    L4C.write(55);

    delay(80);

    L1F.write(70);
    L4F.write(70);

```

```
    delay(50);

    // Other diagonal pair
    L2F.write(140);
    L3F.write(140);
    delay(50);

    L2C.write(95);
    L3C.write(95);

    delay(80);

    L2F.write(120);
    L3F.write(120);
    delay(50);

    stand();
}
void salute()
{
    // Lift Front Right Leg

    L2F.write(150);    // Raise femur
    L2T.write(0);      // Fold tibia

    delay(400);

    // Salute motion
    for(int i = 0; i < 2; i++)
    {
        L2C.write(90);
        delay(250);

        L2C.write(130);
        delay(250);
    }

    // Return to stand position
```

```
L2C.write(110);
L2F.write(120);
L2T.write(120);

delay(300);

stand();
}

// =====
// SETUP
// =====
void setup()
{

    BT.begin(38400);
    Dabble.begin(BT);
    Serial.begin(9600);
    pinMode(trigPin, OUTPUT);
    pinMode(echoPin, INPUT);

    L1C.attach(2);
    L1F.attach(3);
    L1T.attach(4);

    L2C.attach(5);
    L2F.attach(6);
    L2T.attach(7);

    // moved from 8,9,10
    L3C.attach(A0);
    L3F.attach(A1);
    L3T.attach(A2);

    L4C.attach(11);
    L4F.attach(12);
    L4T.attach(13);

    stand();
}
```

```

}

// =====
// LOOP
// =====
void loop()
{
    Dabble.processInput();
    // Enable obstacle mode
    if(GamePad.isCirclePressed())
    {
        obstacleMode = true;
    }

    // Disable obstacle mode
    if(GamePad.isCrossPressed())
    {
        obstacleMode = false;
    }

    // Obstacle Avoidance
    if(obstacleMode)
    {
        float d = getDistance();

        if(d < 20)
        {
            backwardStep();
            delay(100);

            turnRight();
            delay(100);
        }
        else
        {
            forwardStep();
        }

        return;
    }

```



```
}  
  
// Manual Control  
  
if(GamePad.isUpPressed())  
{  
    forwardStep();  
}  
  
else if(GamePad.isDownPressed())  
{  
    backwardStep();  
}  
  
else if(GamePad.isLeftPressed())  
{  
    turnLeft();  
}  
  
else if(GamePad.isRightPressed())  
{  
    turnRight();  
}  
  
else if(GamePad.isSquarePressed())  
{  
    stand();  
}  
if(GamePad.isTrianglePressed())  
{  
    salute();  
}  
}
```